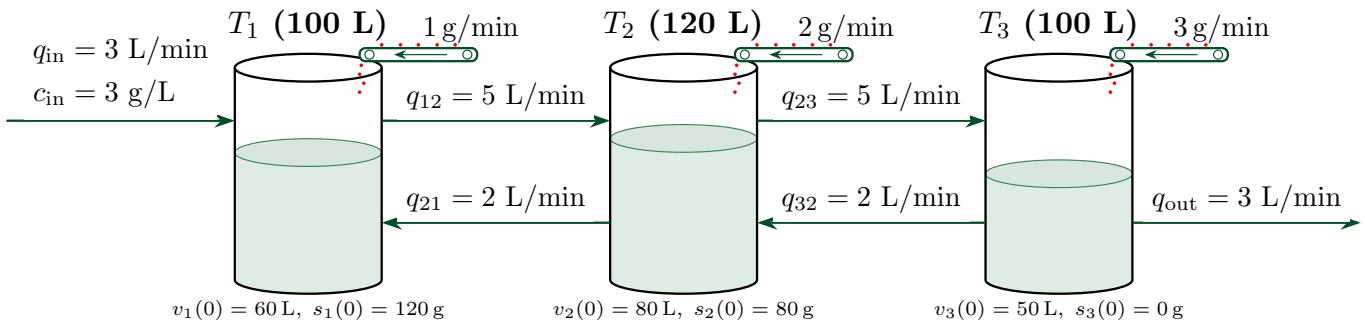
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MA3521 — In-Class Tank Worksheet

Name:

1.



Tanks are well-mixed, t in min, $\mathbf{v}(t)$ in L, $\mathbf{s}(t)$ in g, and conveyers add dry salt at the rate shown.

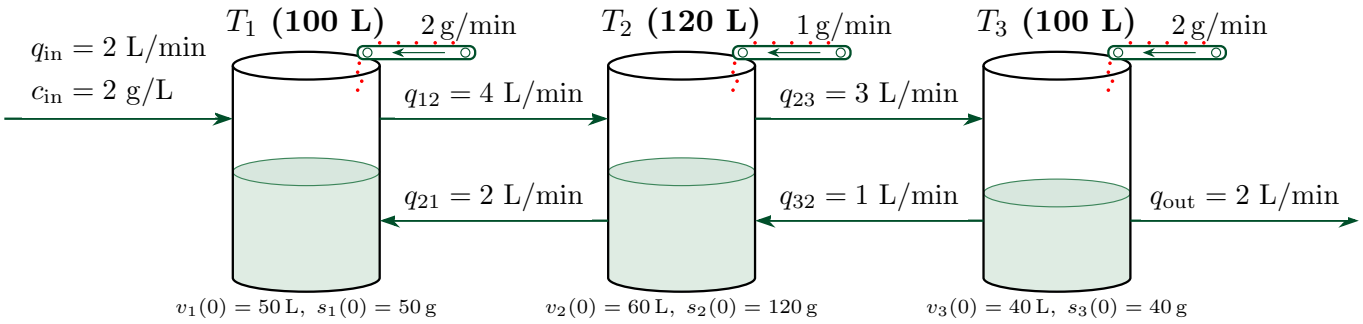
(a) $\mathbf{s}' = \boldsymbol{A}\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \begin{pmatrix} 120 \\ 80 \\ 0 \end{pmatrix}$ g $\quad 0 < t <$ [ ] min

$\mathbf{v}(t) =$ [ ] L $\qquad \mathbf{b} =$ [ ] g/min

(b) $\boldsymbol{A} =$ [ ] min^{-1}

(c) A UC form is $\mathbf{s}_p(t) =$ [ ]. Constants are determined by row reducing $\boldsymbol{M}_{\text{UC}} =$ [ ]

2.



Tanks are well-mixed, t in min, $\mathbf{v}(t)$ in L, $\mathbf{s}(t)$ in g, and conveyers add dry salt at the rate shown.

(a) $\mathbf{s}' = \boldsymbol{A}\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \begin{pmatrix} 50 \\ 120 \\ 40 \end{pmatrix}$ g $\quad 0 < t <$ $\boxed{\infty}$ min

$\mathbf{v}(t) =$ $\boxed{\begin{pmatrix} 50 \\ 60 \\ 40 \end{pmatrix}}$ L $\qquad \mathbf{b} =$ $\boxed{}$ g/min

(b) $\boldsymbol{A} =$ $\boxed{}$ min^{-1}

(c) A UC form is $\mathbf{s}_p(t) =$ $\boxed{}$. Constants are determined by row reducing $\boldsymbol{M}_{\text{UC}} =$ $\boxed{}$

(d) Not on exam. Solve the IVP $\mathbf{y} =$ $\boxed{}$